

- `yum update -y`

```
root@ip-172-31-19-95~$ sudo -i
[root@ip-172-31-19-95 ~]# yum update -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
amzn2-core | 3.6 kB 00:00:00
Resolving Dependencies
--> Running transaction check
--> Package jbigkit-libs.x86_64 0:2.0-11.amzn2.0.2 will be updated
--> Package jbigkit-libs.x86_64 0:2.0-11.amzn2.0.3 will be an update
--> Package kernel.x86_64 0:5.10.197-186.748.amzn2 will be installed
--> Package libsepol.x86_64 0:2.5-8.1.amzn2.0.2 will be updated
--> Package libsepol.x86_64 0:2.5-10.amzn2.0.1 will be an update
--> Package libtiff.x86_64 0:4.0.3-35.amzn2.0.16 will be updated
--> Package libtiff.x86_64 0:4.0.3-35.amzn2.0.19 will be an update
--> Package sudo.x86_64 0:1.8.23-10.amzn2.3.4 will be updated
--> Package sudo.x86_64 0:1.8.23-10.amzn2.3.5 will be an update
--> Package vim-common.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-common.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-data.noarch 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-data.noarch 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-enhanced.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-enhanced.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-filesystem.noarch 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-filesystem.noarch 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-minimal.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-minimal.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package xxd.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package xxd.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package Arch Version Repository Size
=====
Installing:
kernel x86_64 5.10.197-186.748.amzn2 amzn2extra-kernel-5.10 33 M
Updating:
jbigkit-libs x86_64 2.0-11.amzn2.0.3 amzn2-core 47 k
libsepol x86_64 2.5-10.amzn2.0.1 amzn2-core 292 k
=====

Cleanup : 2:vim-enhanced-9.0.1882-1.amzn2.0.1.x86_64 15/21
Cleanup : 2:vim-common-9.0.1882-1.amzn2.0.1.x86_64 16/21
Cleanup : 2:vim-data-9.0.1882-1.amzn2.0.1.noarch 17/21
Cleanup : 2:vim-filesystem-9.0.1882-1.amzn2.0.1.noarch 18/21
Cleanup : 2:xxd-9.0.1882-1.amzn2.0.1.x86_64 19/21
Cleanup : jbigkit-libs-2.0-11.amzn2.0.2.x86_64 20/21
Cleanup : libsepol-2.5-8.1.amzn2.0.2.x86_64 21/21
Verifying : 2:vim-enhanced-9.0.1882-1.amzn2.0.2.x86_64 1/21
Verifying : kernel-5.10.197-186.748.amzn2.x86_64 2/21
Verifying : libsepol-2.5-10.amzn2.0.1.x86_64 3/21
Verifying : 2:xxd-9.0.1882-1.amzn2.0.2.x86_64 4/21
Verifying : libtiff-4.0.3-35.amzn2.0.19.x86_64 5/21
Verifying : 2:vim-minimal-9.0.1882-1.amzn2.0.2.x86_64 6/21
Verifying : 2:vim-common-9.0.1882-1.amzn2.0.2.x86_64 7/21
Verifying : 2:vim-data-9.0.1882-1.amzn2.0.2.noarch 8/21
Verifying : sudo-1.8.23-10.amzn2.3.5.x86_64 9/21
Verifying : jbigkit-libs-2.0-11.amzn2.0.3.x86_64 10/21
Verifying : 2:vim-filesystem-9.0.1882-1.amzn2.0.2.noarch 11/21
Verifying : 2:vim-minimal-9.0.1882-1.amzn2.0.1.x86_64 12/21
Verifying : 2:xxd-9.0.1882-1.amzn2.0.1.x86_64 13/21
Verifying : libtiff-4.0.3-35.amzn2.0.16.x86_64 14/21
Verifying : sudo-1.8.23-10.amzn2.3.4.x86_64 15/21
Verifying : 2:vim-filesystem-9.0.1882-1.amzn2.0.1.noarch 16/21
Verifying : 2:vim-data-9.0.1882-1.amzn2.0.1.noarch 17/21
Verifying : 2:vim-common-9.0.1882-1.amzn2.0.1.x86_64 18/21
Verifying : libsepol-2.5-8.1.amzn2.0.2.x86_64 19/21
Verifying : 2:vim-enhanced-9.0.1882-1.amzn2.0.1.x86_64 20/21
Verifying : jbigkit-libs-2.0-11.amzn2.0.2.x86_64 21/21

Installed:
kernel.x86_64 0:5.10.197-186.748.amzn2

Updated:
jbigkit-libs.x86_64 0:2.0-11.amzn2.0.3 libsepol.x86_64 0:2.5-10.amzn2.0.1 libtiff.x86_64 0:4.0.3-35.amzn2.0.19
sudo.x86_64 0:1.8.23-10.amzn2.3.5 vim-common.x86_64 2:9.0.1882-1.amzn2.0.2 vim-data.noarch 2:9.0.1882-1.amzn2.0.2
vim-enhanced.x86_64 2:9.0.1882-1.amzn2.0.2 vim-filesystem.noarch 2:9.0.1882-1.amzn2.0.2 vim-minimal.x86_64 2:9.0.1882-1.amzn2.0.2
xxd.x86_64 2:9.0.1882-1.amzn2.0.2

Complete!
```

- Yum install httpd -y

```
[root@ip-172-31-19-95 ~]# yum install httpd -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
Resolving Dependencies
--> Running transaction check
--> Package httpd.x86_64 0:2.4.57-1.amzn2 will be installed
--> Processing Dependency: httpd-filesystem = 2.4.57-1.amzn2 for package: httpd-2.4.57-1.amzn2.x86_64
--> Processing Dependency: httpd-tools = 2.4.57-1.amzn2 for package: httpd-2.4.57-1.amzn2.x86_64
--> Processing Dependency: /etc/mime.types for package: httpd-2.4.57-1.amzn2.x86_64
--> Processing Dependency: httpd-filesystem for package: httpd-2.4.57-1.amzn2.x86_64
--> Processing Dependency: mod_http2 for package: httpd-2.4.57-1.amzn2.x86_64
--> Processing Dependency: system-logos-httpd for package: httpd-2.4.57-1.amzn2.x86_64
--> Processing Dependency: libapr-1.so.0(64bit) for package: httpd-2.4.57-1.amzn2.x86_64
--> Processing Dependency: libaprutil-1.so.0(64bit) for package: httpd-2.4.57-1.amzn2.x86_64
--> Running transaction check
--> Package apr.x86_64 0:1.7.2-1.amzn2 will be installed
--> Package apr-util.x86_64 0:1.6.3-1.amzn2.0.1 will be installed
--> Processing Dependency: apr-util-bdb(x86-64) = 1.6.3-1.amzn2.0.1 for package: apr-util-1.6.3-1.amzn2.0.1.x86_64
--> Package generic-logos-httpd.noarch 0:18.0.0-4.amzn2 will be installed
--> Package httpd-filesystem.noarch 0:2.4.57-1.amzn2 will be installed
--> Package httpd-tools.x86_64 0:2.4.57-1.amzn2 will be installed
--> Package mailcap.noarch 0:2.1.41-2.amzn2 will be installed
--> Package mod_http2.x86_64 0:1.15.19-1.amzn2.0.1 will be installed
--> Running transaction check
--> Package apr-util-bdb.x86_64 0:1.6.3-1.amzn2.0.1 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package Arch Version Repository Size
=====
Installing:
httpd x86_64 2.4.57-1.amzn2 amzn2-core 1.4 M
Installing for dependencies:
apr x86_64 1.7.2-1.amzn2 amzn2-core 130 k
apr-util x86_64 1.6.3-1.amzn2.0.1 amzn2-core 101 k
apr-util-bdb x86_64 1.6.3-1.amzn2.0.1 amzn2-core 22 k
generic-logos-httpd noarch 18.0.0-4.amzn2 amzn2-core 19 k
httpd-filesystem noarch 2.4.57-1.amzn2 amzn2-core 24 k
=====

(6/9): apr-util-bdb-1.6.3-1.amzn2.0.1.x86_64.rpm | 22 kB 00:00:00
(7/9): mailcap-2.1.41-2.amzn2.noarch.rpm | 31 kB 00:00:00
(8/9): httpd-tools-2.4.57-1.amzn2.x86_64.rpm | 88 kB 00:00:00
(9/9): mod_http2-1.15.19-1.amzn2.0.1.x86_64.rpm | 149 kB 00:00:00
-----
Total | 6.6 MB/s | 1.9 MB 00:00:00
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Installing : apr-1.7.2-1.amzn2.x86_64 1/9
Installing : apr-util-1.6.3-1.amzn2.0.1.x86_64 2/9
Installing : apr-util-bdb-1.6.3-1.amzn2.0.1.x86_64 3/9
Installing : httpd-tools-2.4.57-1.amzn2.x86_64 4/9
Installing : httpd-filesystem-2.4.57-1.amzn2.noarch 5/9
Installing : generic-logos-httpd-18.0.0-4.amzn2.noarch 6/9
Installing : mailcap-2.1.41-2.amzn2.noarch 7/9
Installing : mod_http2-1.15.19-1.amzn2.0.1.x86_64 8/9
Installing : httpd-2.4.57-1.amzn2.x86_64 9/9
Verifying : apr-util-bdb-1.6.3-1.amzn2.0.1.x86_64 1/9
Verifying : apr-1.7.2-1.amzn2.x86_64 2/9
Verifying : httpd-tools-2.4.57-1.amzn2.x86_64 3/9
Verifying : apr-util-1.6.3-1.amzn2.0.1.x86_64 4/9
Verifying : mailcap-2.1.41-2.amzn2.noarch 5/9
Verifying : generic-logos-httpd-18.0.0-4.amzn2.noarch 6/9
Verifying : mod_http2-1.15.19-1.amzn2.0.1.x86_64 7/9
Verifying : httpd-2.4.57-1.amzn2.x86_64 8/9
Verifying : httpd-filesystem-2.4.57-1.amzn2.noarch 9/9

Installed:
httpd.x86_64 0:2.4.57-1.amzn2

Dependency Installed:
apr.x86_64 0:1.7.2-1.amzn2 apr-util.x86_64 0:1.6.3-1.amzn2.0.1 apr-util-bdb.x86_64 0:1.6.3-1.amzn2.0.1
generic-logos-httpd.noarch 0:18.0.0-4.amzn2 httpd-filesystem.noarch 0:2.4.57-1.amzn2 httpd-tools.x86_64 0:2.4.57-1.amzn2
mailcap.noarch 0:2.1.41-2.amzn2 mod_http2.x86_64 0:1.15.19-1.amzn2.0.1

Complete!
[root@ip-172-31-19-95 ~]#
```

- Vi /etc/hosts
- Add HA-Proxy-Server Pubic IP Address

```
Complete!
[root@ip-172-31-19-95 ~]# vi /etc/hosts
[root@ip-172-31-19-95 ~]# cat /etc/hosts
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost6 localhost6.localdomain6
3.85.10.182 load_balancer
[root@ip-172-31-19-95 ~]#
```

```
root@ip-172-31-19-95 ~#
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost6 localhost6.localdomain6
3.85.10.182 load_balancer|
```

Run below command on Server-1

ping load_balancer -c 4

```
[root@ip-172-31-19-95 ~]# ping load_balancer -c 4
PING load_balancer (3.85.10.182) 56(84) bytes of data.

--- load_balancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3055ms

[root@ip-172-31-19-95 ~]#
```

- systemctl start httpd
- systemctl status httpd

```
[root@ip-172-31-19-95 ~]# systemctl start httpd
[root@ip-172-31-19-95 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; vendor preset: disabled)
   Active: active (running) since Sat 2023-10-21 15:05:39 UTC; 10s ago
     Docs: man:httpd.service(8)
   Main PID: 6683 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
   CGroup: /system.slice/httpd.service
           └─6683 /usr/sbin/httpd -DFOREGROUND
             └─6684 /usr/sbin/httpd -DFOREGROUND
               └─6685 /usr/sbin/httpd -DFOREGROUND
                 └─6686 /usr/sbin/httpd -DFOREGROUND
                   └─6687 /usr/sbin/httpd -DFOREGROUND
                     └─6688 /usr/sbin/httpd -DFOREGROUND

Oct 21 15:05:39 ip-172-31-19-95.ec2.internal systemd[1]: Starting The Apache HTTP Server...
Oct 21 15:05:39 ip-172-31-19-95.ec2.internal systemd[1]: Started The Apache HTTP Server.
[root@ip-172-31-19-95 ~]#
```

Browse with Server-1 Public IP address:80 it will work

Server-2 Steps:

Run following command to Access Server-2

- ```
root@ip-172-31-31-75:~
ADMIN@DESKTOP-KBQTHU6 MINGW64 ~/Downloads
$ ssh -i saddamnewkey.pem ec2-user@54.221.154.94
The authenticity of host '54.221.154.94 (54.221.154.94)' can't be established.
ED25519 key fingerprint is SHA256:SOAxdh+ILTBd2Rr0n+mfv7QP1+tHVvCiHsYE3LX0n/0.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.221.154.94' (ED25519) to the list of known hosts.

#_ Amazon Linux 2
~\#####
~~\#####\
~~\###|
~~\#/AL2 End of Life is 2025-06-30.
~~V~'->
~~~~A newer version of Amazon Linux is available!  
~-.-/  
~/m/'-/_Amazon Linux 2023, GA and supported until 2028-03-15.  
https://aws.amazon.com/linux/amazon-linux-2023/  
  
11 package(s) needed for security, out of 11 available  
Run "sudo yum update" to apply all updates.  
[ec2-user@ip-172-31-31-75 ~]$ sudo -i  
[root@ip-172-31-31-75 ~]#
```

- ```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.221.154.94' (ED25519) to the list of known hosts.
#_
~\_##### Amazon Linux 2
~~\#####\
~~\###| AL2 End of Life is 2025-06-30.
~~\#/
~~V~'-'>
~~~
~~~_./
~~~_/_/
~/m/'_/_/
A newer version of Amazon Linux is available!
Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/
11 package(s) needed for security, out of 11 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-172-31-31-75 ~]$ sudo -i
[root@ip-172-31-31-75 ~]#
```

```
root@ip-172-31-31-75 ~]$ sudo -i
[root@ip-172-31-31-75 ~]# yum update -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
amzn2-core | 3.6 kB 00:00:00
Resolving Dependencies
--> Running transaction check
--> Package jbigkit-libs.x86_64 0:2.0-11.amzn2.0.2 will be updated
--> Package jbigkit-libs.x86_64 0:2.0-11.amzn2.0.3 will be an update
--> Package kernel.x86_64 0:5.10.197-186.748.amzn2 will be installed
--> Package libsepol.x86_64 0:2.5-8.1.amzn2.0.2 will be updated
--> Package libsepol.x86_64 0:2.5-10.amzn2.0.1 will be an update
--> Package libtiff.x86_64 0:4.0.3-35.amzn2.0.16 will be updated
--> Package libtiff.x86_64 0:4.0.3-35.amzn2.0.19 will be an update
--> Package sudo.x86_64 0:1.8.23-10.amzn2.3.4 will be updated
--> Package sudo.x86_64 0:1.8.23-10.amzn2.3.5 will be an update
--> Package vim-common.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-common.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-data.noarch 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-data.noarch 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-enhanced.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-enhanced.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-filesystem.noarch 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-filesystem.noarch 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-minimal.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-minimal.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package xxd.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package xxd.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package Arch Version Repository Size
=====
Installing:
kernel x86_64 5.10.197-186.748.amzn2 amzn2extra-kernel-5.10 33 M
Updating:
jbigkit-libs x86_64 2.0-11.amzn2.0.3 amzn2-core 47 k
libsepol x86_64 2.5-10.amzn2.0.1 amzn2-core 292 k
=====

Clean up : 2:vim-common-9.0.1882-1.amzn2.0.1.x86_64 16/21
Clean up : 2:vim-data-9.0.1882-1.amzn2.0.1.noarch 17/21
Clean up : 2:vim-filesystem-9.0.1882-1.amzn2.0.1.noarch 18/21
Clean up : 2:xxd-9.0.1882-1.amzn2.0.1.x86_64 19/21
Clean up : jbigkit-libs-2.0-11.amzn2.0.2.x86_64 20/21
Clean up : libsepol-2.5-8.1.amzn2.0.2.x86_64 21/21
Verifying : 2:vim-enhanced-9.0.1882-1.amzn2.0.2.x86_64 1/21
Verifying : kernel-5.10.197-186.748.amzn2.x86_64 2/21
Verifying : libsepol-2.5-10.amzn2.0.1.x86_64 3/21
Verifying : 2:xxd-9.0.1882-1.amzn2.0.2.x86_64 4/21
Verifying : libtiff-4.0.3-35.amzn2.0.19.x86_64 5/21
Verifying : 2:vim-minimal-9.0.1882-1.amzn2.0.2.x86_64 6/21
Verifying : 2:vim-common-9.0.1882-1.amzn2.0.2.x86_64 7/21
Verifying : 2:vim-data-9.0.1882-1.amzn2.0.2.noarch 8/21
Verifying : sudo-1.8.23-10.amzn2.3.5.x86_64 9/21
Verifying : jbigkit-libs-2.0-11.amzn2.0.3.x86_64 10/21
Verifying : 2:vim-filesystem-9.0.1882-1.amzn2.0.2.noarch 11/21
Verifying : 2:vim-minimal-9.0.1882-1.amzn2.0.1.x86_64 12/21
Verifying : 2:xxd-9.0.1882-1.amzn2.0.1.x86_64 13/21
Verifying : libtiff-4.0.3-35.amzn2.0.16.x86_64 14/21
Verifying : sudo-1.8.23-10.amzn2.3.4.x86_64 15/21
Verifying : 2:vim-filesystem-9.0.1882-1.amzn2.0.1.noarch 16/21
Verifying : 2:vim-data-9.0.1882-1.amzn2.0.1.noarch 17/21
Verifying : 2:vim-common-9.0.1882-1.amzn2.0.1.x86_64 18/21
Verifying : libsepol-2.5-8.1.amzn2.0.2.x86_64 19/21
Verifying : 2:vim-enhanced-9.0.1882-1.amzn2.0.1.x86_64 20/21
Verifying : jbigkit-libs-2.0-11.amzn2.0.2.x86_64 21/21

Installed:
kernel.x86_64 0:5.10.197-186.748.amzn2

Updated:
jbigkit-libs.x86_64 0:2.0-11.amzn2.0.3 libsepol.x86_64 0:2.5-10.amzn2.0.1 libtiff.x86_64 0:4.0.3-35.amzn2.0.19
sudo.x86_64 0:1.8.23-10.amzn2.3.5 vim-common.x86_64 2:9.0.1882-1.amzn2.0.2 vim-data.noarch 2:9.0.1882-1.amzn2.0.2
vim-enhanced.x86_64 2:9.0.1882-1.amzn2.0.2 vim-filesystem.noarch 2:9.0.1882-1.amzn2.0.2 vim-minimal.x86_64 2:9.0.1882-1.amzn2.0.2
xxd.x86_64 2:9.0.1882-1.amzn2.0.2

Complete!
[root@ip-172-31-31-75 ~]#
```

- yum install nginx -y
- sudo amazon-linux-extras install nginx1 -y



```
Complete!
[root@ip-172-31-31-75 ~]# yum install nginx -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
No package nginx available.
Error: Nothing to do
```

nginx is available in Amazon Linux Extra topic "nginx1"

To use, run

```
sudo amazon-linux-extras install nginx1
```

Learn more at

[https://aws.amazon.com/amazon-linux-2/faqs/#Amazon\\_Linux\\_Extras](https://aws.amazon.com/amazon-linux-2/faqs/#Amazon_Linux_Extras)

```
[root@ip-172-31-31-75 ~]#
```

```
[root@ip-172-31-31-75 ~]# sudo amazon-linux-extras install nginx1 -y
Installing nginx
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
Cleaning repos: amzn2-core amzn2extra-docker amzn2extra-nginx1
17 metadata files removed
6 sqlite files removed
0 metadata files removed
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
amzn2-core | 3.6 kB 00:00:00
amzn2extra-docker | 2.9 kB 00:00:00
amzn2extra-nginx1 | 3.0 kB 00:00:00
amzn2extra-nginx1 | 2.9 kB 00:00:00
(1/9): amzn2-core/2/x86_64/group_gz | 2.7 kB 00:00:00
(2/9): amzn2-core/2/x86_64/updateinfo | 729 kB 00:00:00
(3/9): amzn2extra-docker/2/x86_64/primary_db | 101 kB 00:00:00
(4/9): amzn2extra-nginx1/2/x86_64/updateinfo | 3.0 kB 00:00:00
(5/9): amzn2extra-nginx1/2/x86_64/primary_db | 55 kB 00:00:00
(6/9): amzn2extra-docker/2/x86_64/updateinfo | 13 kB 00:00:00
(7/9): amzn2extra-nginx1-5.10/2/x86_64/updateinfo | 40 kB 00:00:00
(8/9): amzn2extra-nginx1-5.10/2/x86_64/primary_db | 20 MB 00:00:00
(9/9): amzn2-core/2/x86_64/primary_db | 67 MB 00:00:00
Resolving Dependencies
--> Running transaction check
--> Package nginx.x86_64 1:1.22.1-1.amzn2.0.3 will be installed
--> Processing Dependency: nginx-core = 1:1.22.1-1.amzn2.0.3 for package: 1:nginx-1.22.1-1.amzn2.0.3.x86_64
--> Processing Dependency: nginxfilesystem = 1:1.22.1-1.amzn2.0.3 for package: 1:nginx-1.22.1-1.amzn2.0.3.x86_64
--> Running transaction check
--> Package nginx-core.x86_64 1:1.22.1-1.amzn2.0.3 will be installed
--> Processing Dependency: libcrypto.so.1.1(OPENSSL_1_1_0)(64bit) for package: 1:nginx-core-1.22.1-1.amzn2.0.3.x86_64
--> Processing Dependency: libssl.so.1.1(OPENSSL_1_1_0)(64bit) for package: 1:nginx-core-1.22.1-1.amzn2.0.3.x86_64
--> Processing Dependency: libssl.so.1.1(OPENSSL_1_1_1)(64bit) for package: 1:nginx-core-1.22.1-1.amzn2.0.3.x86_64
--> Processing Dependency: libcrypto.so.1.1(64bit) for package: 1:nginx-core-1.22.1-1.amzn2.0.3.x86_64
--> Processing Dependency: libprofiler.so.0(64bit) for package: 1:nginx-core-1.22.1-1.amzn2.0.3.x86_64
--> Processing Dependency: libssl.so.1.1(64bit) for package: 1:nginx-core-1.22.1-1.amzn2.0.3.x86_64
--> Package nginxfilesystem.noarch 1:1.22.1-1.amzn2.0.3 will be installed
--> Running transaction check
--> Package gperftools-libs.x86_64 0:2.6.1-1.amzn2 will be installed
--> Package openssl11-libs.x86_64 1:1.1.1g-12.amzn2.0.18 will be installed
```

- vi /etc/hosts
- Add HA-Proxy-Server Pubic IP Address

```
[root@ip-172-31-31-75 ~]# vi /etc/hosts
[root@ip-172-31-31-75 ~]# cat /etc/hosts
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost6 localhost6.localdomain6

3.85.10.182 load_balancer
[root@ip-172-31-31-75 ~]#
```

```
root@ip-172-31-31-75:~#
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost6 localhost6.localdomain6
3.85.10.182 load_balancer
-- INSERT --
4,17 All
```

Run below command on Server-2

ping load\_balancer -c 4

```
[root@ip-172-31-31-75 ~]# ping load_balancer -c 4
PING load_balancer (3.85.10.182) 56(84) bytes of data.

--- load_balancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3053ms

[root@ip-172-31-31-75 ~]#
```

- systemctl start nginx
- systemctl status nginx

```
[root@ip-172-31-31-75 ~]# systemctl start nginx
[root@ip-172-31-31-75 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
 Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; vendor preset: disabled)
 Active: active (running) since Sat 2023-10-21 15:09:49 UTC; 5s ago
 Process: 6739 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
 Process: 6735 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
 Process: 6734 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
 Main PID: 6741 (nginx)
 CGroup: /system.slice/nginx.service
 └─6741 nginx: master process /usr/sbin/nginx
 └─6742 nginx: worker process

Oct 21 15:09:49 ip-172-31-31-75.ec2.internal systemd[1]: Starting The nginx HTTP and reverse proxy server...
Oct 21 15:09:49 ip-172-31-31-75.ec2.internal nginx[6735]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Oct 21 15:09:49 ip-172-31-31-75.ec2.internal nginx[6735]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Oct 21 15:09:49 ip-172-31-31-75.ec2.internal systemd[1]: Started The nginx HTTP and reverse proxy server.
[root@ip-172-31-31-75 ~]#
```

Browse with Server-2 Public IP address:80 it will work

## HA-Proxy-Server Steps:

Run following command to Access HA-Proy-Server

- ssh -i your-key.pem ec2-user@Pubipaddress of HA-Proxy-Server instance





```
root@ip-172-31-26-203~# yum update -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
amzn2-core | 3.6 kB 00:00:00
Resolving Dependencies
--> Running transaction check
--> Package jbigkit-libs.x86_64 0:2.0-11.amzn2.0.2 will be updated
--> Package jbigkit-libs.x86_64 0:2.0-11.amzn2.0.3 will be an update
--> Package kernel.x86_64 0:5.10.197-186.748.amzn2 will be installed
--> Package libsepol.x86_64 0:2.5-8.1.amzn2.0.2 will be updated
--> Package libsepol.x86_64 0:2.5-10.amzn2.0.1 will be an update
--> Package libtiff.x86_64 0:4.0.3-35.amzn2.0.16 will be updated
--> Package libtiff.x86_64 0:4.0.3-35.amzn2.0.19 will be an update
--> Package sudo.x86_64 0:1.8.23-10.amzn2.3.4 will be updated
--> Package sudo.x86_64 0:1.8.23-10.amzn2.3.5 will be an update
--> Package vim-common.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-common.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-data.noarch 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-data.noarch 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-enhanced.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-enhanced.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-filesystem.noarch 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-filesystem.noarch 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package vim-minimal.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package vim-minimal.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Package xxd.x86_64 2:9.0.1882-1.amzn2.0.1 will be updated
--> Package xxd.x86_64 2:9.0.1882-1.amzn2.0.2 will be an update
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package Arch Version Repository Size
=====
Installing:
kernel x86_64 5.10.197-186.748.amzn2 amzn2extra-kernel-5.10 33 M
Updating:
jbigkit-libs x86_64 2.0-11.amzn2.0.3 amzn2-core 47 k
libsepol x86_64 2.5-10.amzn2.0.1 amzn2-core 292 k
libtiff x86_64 4.0.3-35.amzn2.0.19 amzn2-core 175 k
=====
```

```
root@ip-172-31-26-203~#
Cleanup : 2:vim-common-9.0.1882-1.amzn2.0.1.x86_64 16/21
Cleanup : 2:vim-data-9.0.1882-1.amzn2.0.1.noarch 17/21
Cleanup : 2:vim-filesystem-9.0.1882-1.amzn2.0.1.noarch 18/21
Cleanup : 2:xxd-9.0.1882-1.amzn2.0.1.x86_64 19/21
Cleanup : jbigkit-libs-2.0-11.amzn2.0.2.x86_64 20/21
Cleanup : libsepol-2.5-8.1.amzn2.0.2.x86_64 21/21
Verifying : 2:vim-enhanced-9.0.1882-1.amzn2.0.2.x86_64 1/21
Verifying : kernel-5.10.197-186.748.amzn2.x86_64 2/21
Verifying : libsepol-2.5-10.amzn2.0.1.x86_64 3/21
Verifying : 2:xxd-9.0.1882-1.amzn2.0.2.x86_64 4/21
Verifying : libtiff-4.0.3-35.amzn2.0.19.x86_64 5/21
Verifying : 2:vim-minimal-9.0.1882-1.amzn2.0.2.x86_64 6/21
Verifying : 2:vim-common-9.0.1882-1.amzn2.0.2.x86_64 7/21
Verifying : 2:vim-data-9.0.1882-1.amzn2.0.2.noarch 8/21
Verifying : sudo-1.8.23-10.amzn2.3.5.x86_64 9/21
Verifying : jbigkit-libs-2.0-11.amzn2.0.3.x86_64 10/21
Verifying : 2:vim-filesystem-9.0.1882-1.amzn2.0.2.noarch 11/21
Verifying : 2:vim-minimal-9.0.1882-1.amzn2.0.1.x86_64 12/21
Verifying : 2:xxd-9.0.1882-1.amzn2.0.1.x86_64 13/21
Verifying : libtiff-4.0.3-35.amzn2.0.16.x86_64 14/21
Verifying : sudo-1.8.23-10.amzn2.3.4.x86_64 15/21
Verifying : 2:vim-filesystem-9.0.1882-1.amzn2.0.1.noarch 16/21
Verifying : 2:vim-data-9.0.1882-1.amzn2.0.1.noarch 17/21
Verifying : 2:vim-common-9.0.1882-1.amzn2.0.1.x86_64 18/21
Verifying : libsepol-2.5-8.1.amzn2.0.2.x86_64 19/21
Verifying : 2:vim-enhanced-9.0.1882-1.amzn2.0.1.x86_64 20/21
Verifying : jbigkit-libs-2.0-11.amzn2.0.2.x86_64 21/21

Installed:
kernel.x86_64 0:5.10.197-186.748.amzn2

Updated:
jbigkit-libs.x86_64 0:2.0-11.amzn2.0.3 libsepol.x86_64 0:2.5-10.amzn2.0.1 libtiff.x86_64 0:4.0.3-35.amzn2.0.19
sudo.x86_64 0:1.8.23-10.amzn2.3.5 vim-common.x86_64 2:9.0.1882-1.amzn2.0.2 vim-data.noarch 2:9.0.1882-1.amzn2.0.2
vim-enhanced.x86_64 2:9.0.1882-1.amzn2.0.2 vim-filesystem.noarch 2:9.0.1882-1.amzn2.0.2 vim-minimal.x86_64 2:9.0.1882-1.amzn2.0.2
xxd.x86_64 2:9.0.1882-1.amzn2.0.2

Complete!
[root@ip-172-31-26-203 ~]#
```

- yum install haproxy -y

- vi /etc/hosts
- Add Server-1, Server-2 Public IP's

```
root@ip-172.31.26.203:~
127.0.0.1 localhost localhost.localhost localhost4 localhost4.localhost
::1 localhost6 localhost6.localhost
3.85.92.188 server-1
54.221.154.94 server-2
```

- vi /etc/haproxy/haproxy.cfg
- Add Server-1, Server-2 public IP's

```
root@ip-172-31-26-203:~#
user haproxy
group haproxy
daemon

turn on stats unix socket
stats socket /var/lib/haproxy/stats

#-----
common defaults that all the 'listen' and 'backend' sections will
use if not designated in their block
#-----
defaults
 mode http
 log global
 option httplog
 option dontlognull
 option http-server-close
 option forwardfor except 127.0.0.0/8
 option redispatch
 retries 3
 timeout http-request 10s
 timeout queue 1m
 timeout connect 10s
 timeout client 1m
 timeout server 1m
 timeout http-keep-alive 10s
 timeout check 10s
 maxconn 3000

#-----
main frontend which proxys to the backends
#-----
frontend main *:80
 acl url_static path_beg -i /static /images /javascript /stylesheets
 acl url_static path_end -i .jpg .gif .png .css .js

 use_backend static if url_static
 default_backend app

#-----
static backend for serving up images, stylesheets and such
#-----
backend static
 balance roundrobin
 server static 127.0.0.1:4331 check

#-----
round robin balancing between the various backends
#-----
backend app
 balance roundrobin
 server app1 127.0.0.1:5001 check
 server app2 127.0.0.1:5002 check
 server app3 127.0.0.1:5003 check
 server app4 127.0.0.1:5004 check
 server app5 3.85.92.188:80 check
 server app6 54.221.154.94:80 check
-- INSERT --
```

- systemctl enable haproxy
- systemctl start haproxy
- systemctl status haproxy

```
root@ip-172-31-26-203:~#
[root@ip-172-31-26-203 ~]# systemctl enable haproxy
Created symlink from /etc/systemd/system/multi-user.target.wants/haproxy.service to /usr/lib/systemd/system/haproxy.service.
[root@ip-172-31-26-203 ~]# systemctl start haproxy
[root@ip-172-31-26-203 ~]# systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
 Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; vendor preset: disabled)
 Active: active (running) since Sat 2023-10-21 15:15:45 UTC; 11s ago
 Main PID: 6718 (haproxy-systemd)
 CGroup: /system.slice/haproxy.service
 └─6718 /usr/sbin/haproxy-systemd-wrapper -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid
 └─6719 /usr/sbin/haproxy -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -Ds
 └─6720 /usr/sbin/haproxy -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -Ds

Oct 21 15:15:45 ip-172-31-26-203.ec2.internal systemd[1]: Started HAProxy Load Balancer.
Oct 21 15:15:45 ip-172-31-26-203.ec2.internal haproxy-systemd-wrapper[6718]: haproxy-systemd-wrapper: executing /usr/sbin/haproxy -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -Ds
[root@ip-172-31-26-203 ~]#
```

- now browse with HA-Proxy-Server PublicIP:80 it will distribute load to Server-1, Server-2